

The Bottom Line: Musings on Linearity in Math and in Life

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1 Introduction

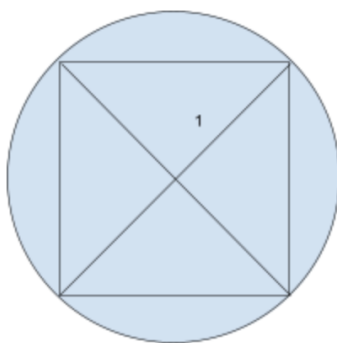
In early April, I sent my math teacher a tweet on behalf of my friend Maxwell that said, “i used to think the intermediate value theorem was overrated; now I think it’s underrated. thus there must have been a point in time when.” My math teacher responded with, “jokes on him. He’s assuming thinking is continuous and I’ve seen plenty of contradictions to that.” After a few years of teaching grubby high school kids, I’m sure he’s seen all kinds of things. At the end of the day, what does math have to do with thinking? If we represent everything with a line, do we also use hearts, squares and triangles? My friend Alyssa on the soccer team probably thinks in truncated icosahedrons when she’s worshipping Marcus Rashford. Though I don’t have an answer to any of the questions mentioned above, I do have some examples of how these pesky lines wiggle into our daily lives.

2 Linear Reasoning

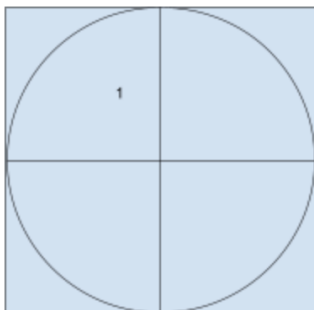
Do you support standardized tests? Well, you probably also want a wall on the Mexico-United States border. Not taking any sides here, but doesn’t this sound like the last Reddit fight you got into? This line of thinking is linear, though not very continuous. Linear reasoning is a systemic style of thinking, where one idea proceeds another like points on a line. It all goes back to fifth century BCE Greece when people were working with whole numbers and messing around with the Pythagorean Theorem. Someone from the Pythagoras school realized that no isosceles right triangle can be represented using three whole number length sides. Therefore, some believed that the hypotenuse of isosceles right triangles and π were simply not numbers. This was pretty mind-boggling, and legends say that Hippasus, the person who proved the square root of 2 is irrational, was drowned in the sea by his colleagues for suggesting the theorem. Luckily, the idea stuck around. Mathematicians like Euclid and Archimedes later understood that measuring with only whole numbers is a thing of the past and were ready to move on. If circles can’t be measured, then how have people been making wheels?

It doesn’t seem very continuous for me to go from whole numbers to calculus, but what better way to find the area of a circle than to use limits? The method of exhaustion finds the area of shapes, such as circles, by inscribing a sequence of polygons inside of it. The areas of these polygons

then converge to the area of the shape. It seems like the idea has been popping up from fifth century BC's Antiphon, then from Eudoxus of Cnidus, and then from Liu Hui in China. Euclid uses the method to prove six propositions in book 12 of the *Elements*, and Archimedes used it to measure the circle. Starting from an inscribed square, we get a pretty easy but bad estimate. If the circle's radius is 1 unit, then the square's area would be 2 units².



Now let's consider a circumscribing square! It's also a pretty bad estimate. With a radius of 1 unit, this square would have an area of 4 units². Knowing that our circle's area is π , we just narrowed down the value of π to between 2 and 4!



What? Not very impressive? We at least know that π is a number and not some unmeasurable wheel! Archimedes did go further than squares though. He bounded π with the numbers $\frac{223}{71}$ and $\frac{210}{70}$ by inscribing and circumscribing 96-gons about the circle. Some of his other findings using the method of exhaustion include:

1. The area bounded by the intersection between a line and a parabola is $\frac{4}{3}$ of the area of a triangle that has the same base and height.

2. The areas of ellipses are in proportion to rectangles with sides that have the same measure as the ellipsis' major axes and minor axes.
3. The volume of a sphere is 4 times the volume of a cone that has a base with the same radius of the sphere and a height equal to the radius of the sphere.
4. The volume of a cylinder with a height that is equal to the diameter of its base is $\frac{3}{2}$ of the volume of a sphere with the same diameter.
5. The area bounded by a spiral rotation and a line is $\frac{1}{3}$ the area of a circle with a radius that has the same measure as the line segment.

The first successful evaluations of infinite series also used the method of exhaustion.

You may be wondering how any of this has to do with linear reasoning, but we're almost there. First off, meet Greg.



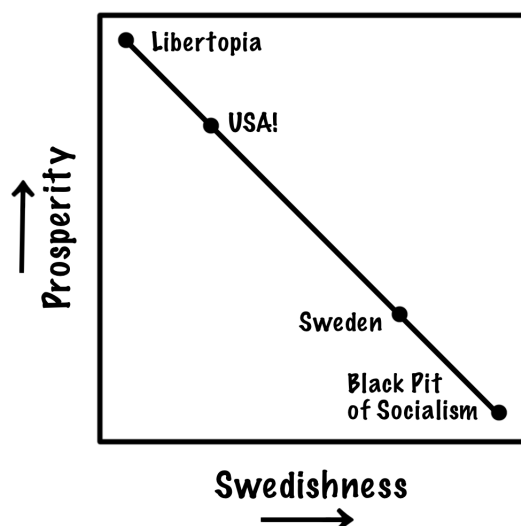
Throughout his childhood, his favorite compliment is that he's "well-rounded." Why may a circle be proud of being round? you ask. Well, that's because Greg fooled them all! He's actually a 325325325325325325-gon! Did the mysterious smile give it away? In calculus, you calculate many things by breaking them up into small lines, or the slope at one point. The more you zoom in to a curve, the straighter it becomes. And this is true for all curves!

Well, this is where linear reasoning comes back into the conversation. Thinking in a systematic way, where one point connects to the next with a straight line seems fair, so why do I still feel funny inside when Reddit_b0i_THEb0sser accused me of being a sus-scrofa-domesticus when I said I just want to eat food and sleep this summer? The reason may be that everything in life does not

always act as lines! We just like to think in lines because on a super small scale, curves are very similar to lines.

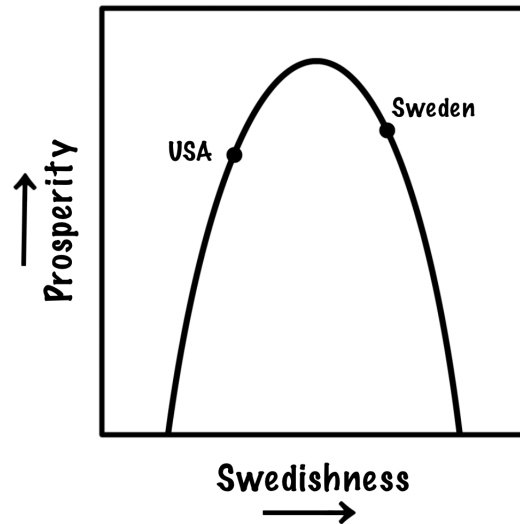
3 Linear Reasoning in Politics

What makes the hottest argument on the Internet? Probably politics. Who completely abuses linear reasoning? Probably political shouters. I do hope that this article receives as much attention as they do, so let's get political. After half of a semester of economics, a good portion of my class now believes that living anywhere beats living in America. Going off of an example from Jordan Ellenberg's book, *How Not to Be Wrong*, I'm sure there's at least one person in my class who is ready to move to Sweden. Ellenberg brings up the battle over the Affordable Care Act, when a title emerged asking, "Why Is Obama Trying to Make American More Like Sweden When Swedes Are Trying to Be Less Like Sweden?" Ellenberg maps the post's line of thought using the diagram below:



At first glance, the title might've gotten you. Yet after a moment of thought, we find many complications, which has to do with our conversation about linear reasoning. Although it seems logical at first, when the scope of the topic gets bigger, we're no longer dealing with lines but with curves.

The graph below suits Ellenberg's taste much better.



Heading towards each other does not have to mean degradation for either one. Instead, both could be heading towards the tip of prosperity! The idea is probably that Sweden taxes their people too much and the USA not enough, so maybe both countries should balance out a bit. I'm not very sure what the situation is with Sweden overall, but hey, we're here for the math anyway. The curve above is also known as the Laffer curve. Named after economist Andrew Laffer, it is the visualization of a theory that deals with how tax rates and the amount of tax collected by the government affect each other. If, like Sweden in this diagram, people get taxed on the higher side, the consequence is that there will be a decreased amount of taxed activities, resulting in less government revenue. The entire theory depends on the assumption that people will change the way they spend or save due to the effects of government services paid by income tax rates.

4 Conclusion

Many lines of writing about lines later, one may or may not be thinking about all the embarrassing times one overused linear reasoning in a battle of words. From whole numbers to Laffer curves, we've gone a long way. In what shapes do you think in?

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